



Arduino + BC547: 3 Simple Practical Circuits

⚠ Important note for students

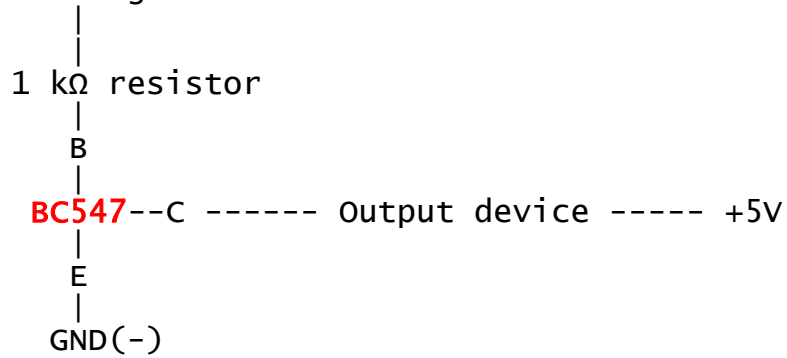
Electronic things you should invest in as a OL & AL teenager. A major subject in ICT revolves around understanding & building small electronic devices around microcontrollers (i.e Arduino boards). The following circuit information is built on O/L electronics & circuit design knowledge.

- Small Soldering Iron & Holder
- Have a small box of electronic/circuit components. Maintain a good understanding of places to purchase these products.
 1. Unitec Trading, 1st Cross Street (Electronic & Arduino boards, etc)
 2. Ethula Building, 1st Cross Street (electronic components, transformers, etc)
- Make space on your table / 6 Square Feet of Empty Table space
- Download the Circuit Design PDF Files.
- Start by building a Full-Wave Rectifier using a small 4.5v Transformer, 4x Diodes & 1000 uF Capacitor.

A **BC547** is a **small NPN transistor** and should be used as an electronic switch for **low-current loads**. A small DC motor can draw more current than a BC547 safely handles, so for the fan project use a **very small low-current motor**, or preferably replace the BC547 with a power transistor/MOSFET if the motor requires more than about 100 mA.

For all three circuits, the basic transistor arrangement is:

Arduino Digital Pin



For a motor, add a **1N4007 diode across the motor** to protect the transistor from back-EMF.

All transistors comprise of a Emitter, Collector & Base.

Circuit 1 — Automatic Temperature-Controlled Fan

Objective

Build a system when:

The temperature becomes too high → Arduino detects it → BC547 switches ON → small fan motor runs.

This is an excellent introductory project because it demonstrates a **real-world automatic control system**. All ICT students should have access to a personal Computer/Laptop. If its possible, get access to the School Computer Lab Computer.

You will notice the that many remote-control toys also employ a diode across the motor. This is to prevent the motor from creating an electro magnetic flux in the reverse direction when its powered down. As you have learned in Grade 11, the DC Electric motor has magnets, and can act like a dynamo when spun.

Power Input vs. Pin Outputs

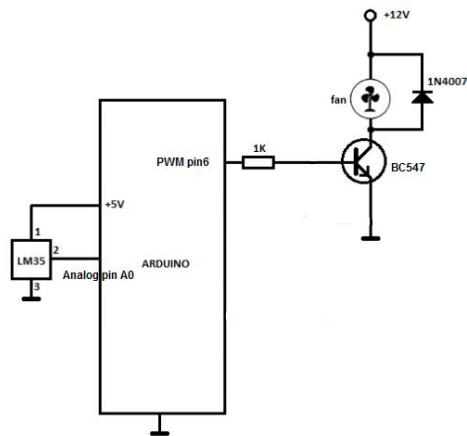
- **Safe Input:** You can safely plug a 12V source into the [Arduino Uno](#) barrel jack or VIN pin. The onboard regulator drops this down to 5V for the chip.
- **Unsafe Pin Voltage:** Never apply 12V directly to the digital or analog pins, and do not try to draw 12V *out* from the 5V or digital pins. The input pins and output logic are strictly 5V max.

Overheating and Current Limits

- **Regulator Heat:** At 12V, the onboard regulator burns off the extra 7V as heat. If you pull too much current from the 5V pin to power extra sensors, LEDs, or motors, the board will overheat.
- **Best Practice:** Keep external loads attached to the board's 5V pin very small. If you need to power 12V components (like motors or relays), use a separate 12V power source and control them via transistors, MOSFETs, or a driver board.

Components

- Arduino Uno
- BC547 transistor
- LM35 temperature sensor
- Small 5V DC motor
- Small fan blade
- 1 k Ω resistor
- 1N4007 diode
- Breadboard
- Jumper wires
- External 5V supply for the motor if required



Connections

LM35

With the flat side of the LM35 facing you:

LM35

Left → Arduino 5V
 Middle → Arduino A0
 Right → Arduino GND

BC547 + Motor

Arduino D8

```

  |
1k $\Omega$ 
  |
  B
BC547 E --> GND
  C
  |
Motor
  |
+5V
  
```

Put the **1N4007** across the motor:

```

+5V
  |
Motor
  |
+-----|<|-----+
  |    1N4007    |
+-----+
  
```

The diode's **cathode/striped end goes toward +5V**.

Arduino program

```
const int tempPin = A0;
const int fanPin = 8;

void setup() {
  pinMode(fanPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {

  int sensorValue = analogRead(tempPin);

  float voltage = sensorValue * (5.0 / 1023.0);

  float temperature = voltage * 100.0;

  Serial.print("Temperature: ");
  Serial.print(temperature);
  Serial.println(" C");

  if (temperature >= 30) {
    digitalWrite(fanPin, HIGH);
  }
  else {
    digitalWrite(fanPin, LOW);
  }

  delay(1000);
}
```

How it works

The LM35 produces an analogue voltage corresponding to temperature.

Arduino performs:

Temperature → Analogue voltage → ADC value → Program decision

For example:

Temperature < 30°C

↓
Fan OFF

Temperature ≥ 30°C

↓
Arduino D8 = HIGH
↓
BC547 ON
↓
Motor ON

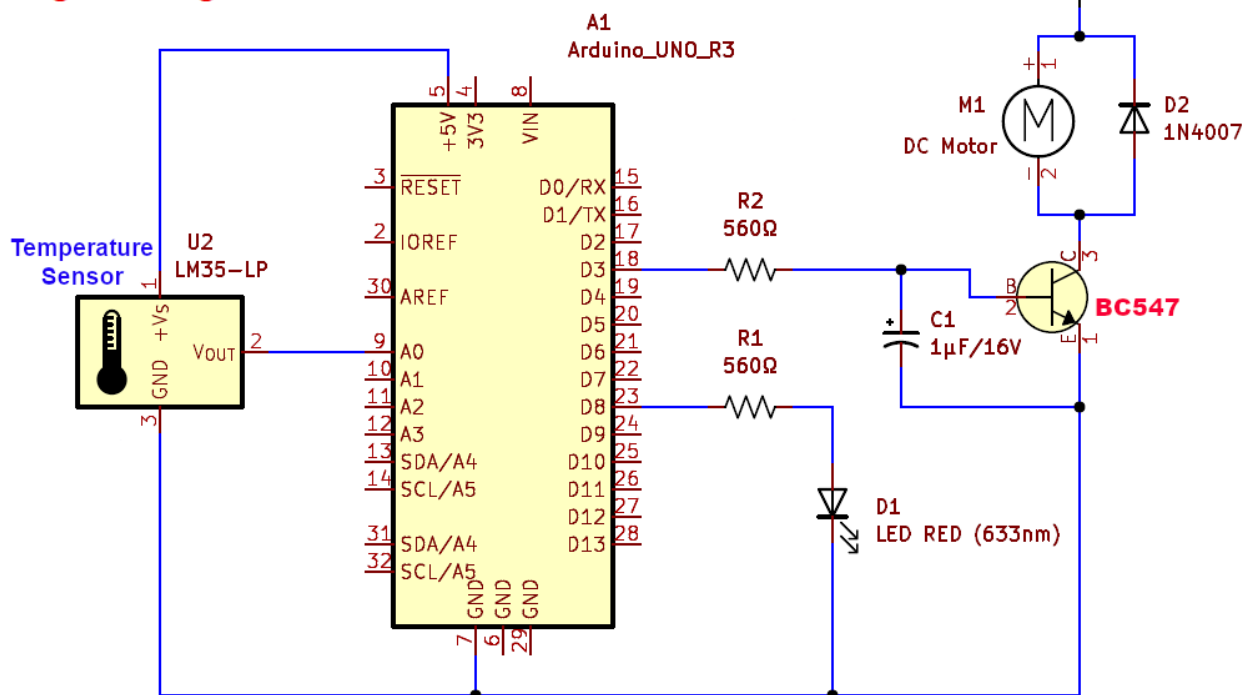
↓
Fan ON

A/L concepts covered

- Sensors
- Analogue input
- ADC
- Digital output
- Conditional statements
- Transistors
- Electronic switching
- Feedback/control systems

A Complete Diagram of DC Motor with LM35 Sensor

Automatic Fan Speed Control Circuit Arduino LM35 Programming



Circuit 2 — Automatic Night Light Using an LDR

Objective

When the environment becomes dark → Arduino detects darkness → BC547 switches ON → LED lamp turns ON.

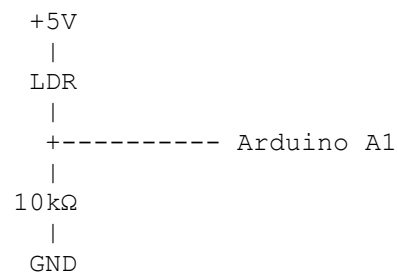
This is simpler than the temperature project and is ideal as the student's first sensor/transistor project.

Components

- Arduino Uno
- BC547
- LDR
- 10 k Ω resistor
- LED
- 220 Ω resistor
- 1 k Ω resistor
- Breadboard
- Jumper wires

LDR circuit

Create a voltage divider:



BC547 + LED

Arduino D7
|
1k Ω
|
B
BC547 -- E $\rightarrow$ GND
C
|
LED
|
220 Ω
|
+5V

Arduino program

```
const int ldrPin = A1;
const int lightPin = 7;

void setup() {
  pinMode(lightPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {

  int lightLevel = analogRead(ldrPin);

  Serial.print("Light Level: ");
  Serial.println(lightLevel);

  if (lightLevel < 400) {
    digitalWrite(lightPin, HIGH);
  }
  else {
    digitalWrite(lightPin, LOW);
  }

  delay(500);
}
```

The exact value 400 will depend on the LDR and room lighting. Students can experiment with the value.

Experiment

Ask students to:

1. Record the LDR value in bright sunlight.
2. Record the value in normal room lighting.
3. Cover the LDR with their hand.
4. Record the value.
5. Determine an appropriate threshold.

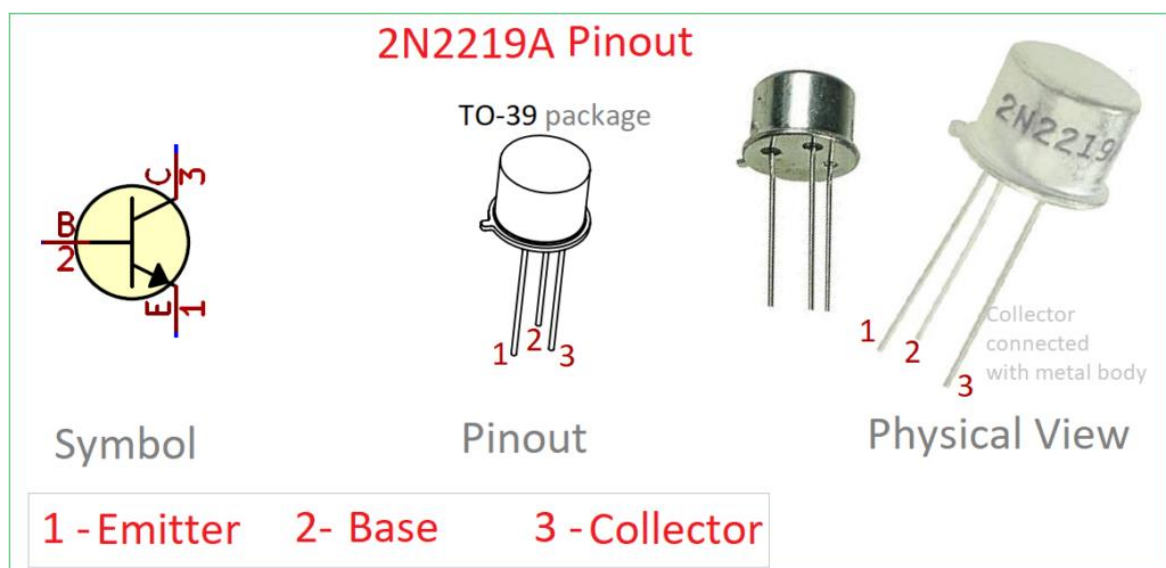
This turns the project into a small **data-analysis exercise**.

A/L concepts covered

- Analogue sensors
 - Voltage divider
 - ADC
 - Threshold values
 - `if...else`
 - Transistor switching
 - Automation
-

As an alternative to the BC547, you can also select the 2N2219A Transistor which has a better voltage handling capacity.

Transistor 2N2219A Pinout



2N2219A is a NPN Switching Silicon Transistor. This is widely used for High frequency Switching Applications. It has good power handling capacity with maximum collector current (I_C) around 800 mA, Collector to base Voltage (VCB) of 75V and Collector to Emitter Voltage (VCE) of 40V. This device comes in TO-39 metal can package. This transistor is ideal for small light-weight electronics, such as motor driven toys, relay-switches, buzzers, small-DC-fans, laptop-speaker-circuits, or actuators.

Circuit 3 — Automatic Soil-Moisture Warning System

For the third project, introduce a **soil-moisture sensor** and use the BC547 to activate a buzzer when the plant becomes too dry.

Objective

If soil moisture falls below the required level → Arduino detects dry soil → BC547 activates buzzer.

This is a nice introduction to an **IoT/Smart Agriculture** concept.

Components

- Arduino Uno
- BC547
- Soil moisture sensor/module
- Active 5V buzzer
- 1 kΩ resistor
- Breadboard
- Jumper wires
- Small plant/pot

Sensor connection

If using a typical soil-moisture module:

Sensor VCC → Arduino 5V
Sensor GND → Arduino GND
Sensor AO → Arduino A2

BC547 + buzzer

```
Arduino D6
|
1kΩ
|
B
BC547 -- E → GND
C
|
Buzzer
|
+5V
```

Arduino program

```
const int moisturePin = A2;
```

```

const int buzzerPin = 6;

void setup() {
  pinMode(buzzerPin, OUTPUT);
  Serial.begin(9600);
}

void loop() {

  int moisture = analogRead(moisturePin);

  Serial.print("Moisture: ");
  Serial.println(moisture);

  if (moisture < 400) {
    digitalWrite(buzzerPin, HIGH);
  }
  else {
    digitalWrite(buzzerPin, LOW);
  }

  delay(1000);
}

```

Important: Moisture sensors/modules can have opposite readings depending on the module. Students should first observe the readings in **dry and wet soil** and then adjust the `< 400` condition accordingly.

Suggested A/L Student Practical Exercise

I would actually turn these into a **3-stage practical assignment**.

Practical	Sensor	Arduino Input	BC547 Output	Programming Concept
 Temperature Fan	LM35	A0	Motor/Fan	<code>if</code> condition
 Automatic Light	LDR	A1	LED	Threshold
 Smart Plant	Moisture Sensor	A2	Buzzer	Sensor decision

Then give students the following challenge:

Level 1 — Basic

Make the circuit work according to the instructions.

Level 2 — Modify

Change the threshold values.

For example:

```
if (temperature >= 30)
```

Change it to:

```
if (temperature >= 35)
```

Observe what happens.

Level 3 — Multiple conditions

For the temperature system:

```
Temperature < 25°C
```

↓

```
Fan OFF
```

```
25°C – 30°C
```

↓

```
Fan LOW
```

```
> 30°C
```

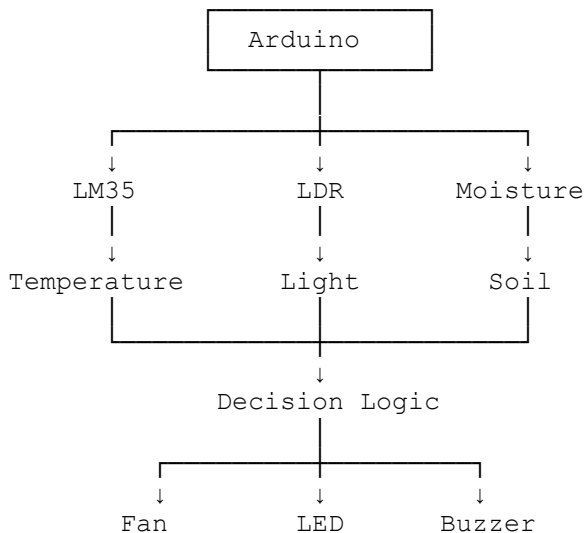
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```
Fan HIGH
```

This introduces students to **multiple if/else conditions and PWM**.

Level 4 — Real-world system

Combine the sensors:



This gives the students a small **Smart Home / Smart Agriculture automation system** and demonstrates how the Arduino acts as the **controller**, while the BC547 acts as an **electronic switch**.

One important teaching point: have students check the **BC547 pinout for the exact manufacturer/package they are using** before wiring it; transistor pin arrangements can differ between parts.

